DBMS Lab Practical 2 – SQL DDL Statements

Problem Statement:

1. Create table Customers with schema (cust_id, cust_name, product, quantity, total_price)
2. Use sequence/ auto-increment for incrementing customer ID and Insert 5 customer records to the table Customers
3. Alter the table Customers by adding one column 'price_per_qnty'
4. Create view 'Cust_View' on Customers displaying customer ID, customer name
5. Update the view 'Cust_View' to display customer ID, product, total price
6. Drop the view 'Cust_View'
7. Create index 'Cust_index' on customer name
8. Drop index 'Cust_index'.
9.Use the name alias for table Customers /Create Synonyms(rename the table in query)
10.Rename the table in query
11.Truncate Table Statement
12.Drop the table Customers

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```
mysql> SHOW TABLES;
```

```
+--------------------+
| Tables_in_Customer |
+--------------------+
| Cust_Synonym       |
| customers          |
+--------------------+
2 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Customers (
      ->    cust_id INT NOT NULL AUTO_INCREMENT,
      ->    cust_name VARCHAR(50),
      ->    product VARCHAR(50),
      ->    quantity INT,
      ->    total_price DECIMAL(10,2),
      ->    PRIMARY KEY (cust_id)
      -> );
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SHOW TABLES;
```

```
+--------------------+
| Tables_in_Customer |
+--------------------+
| Cust_Synonym       |
| Customers          |
| customers          |
+--------------------+
3 rows in set (0.00 sec)
```

```
mysql> DESC Customers;
```

```
+-------------+---------------+------+-----+---------+----------------+
| Field     | Type            | Null | Key | Default | Extra           |
+-------------+---------------+------+-----+---------+----------------+
| cust_id   | int             | NO   | PRI | NULL     | auto_increment |
| cust_name   | varchar(50)   | YES  |    | NULL      |                 |
| product   | varchar(50)   | YES  |      | NULL      |                 |
| quantity  | int             | YES  |    | NULL      |                 |
| total_price | decimal(10,2) | YES  |    | NULL      |                 |
+-------------+---------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Customers;
```

```
Empty set (0.00 sec)
```

```
mysql> INSERT INTO Customers (cust_name, product, quantity, total_price) VALUES
      -> ('Alice', 'Laptop', 2, 2000.00),
      -> ('Bob', 'Phone', 3, 1500.00),
      -> ('Charlie', 'Tablet', 1, 500.00),
      -> ('Diana', 'Monitor', 2, 400.00),
      -> ('Ethan', 'Keyboard', 5, 250.00);
```

```
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Customers;
```

```
+---------+-----------+----------+----------+-------------+
| cust_id | cust_name | product  | quantity | total_price |
+---------+-----------+----------+----------+-------------+
|     1 | Alice   | Laptop   |      2 |   2000.00 |
|     2 | Bob     | Phone     |     3 |   1500.00 |
|     3 | Charlie   | Tablet   |          1 |   500.00 |
|     4 | Diana   | Monitor  |      2 |   400.00 |
|     5 | Ethan   | Keyboard |      5 |   250.00 |
+---------+-----------+----------+----------+-------------+
5 rows in set (0.00 sec)
```

```
mysql> -- Alter table to add price_per_qnty
mysql> SHOW COLUMNS FROM Customers;
```

```
+-------------+---------------+------+-----+---------+----------------+
| Field     | Type            | Null | Key | Default | Extra           |
+-------------+---------------+------+-----+---------+----------------+
| cust_id   | int             | NO   | PRI | NULL     | auto_increment |
| cust_name   | varchar(50)   | YES  |    | NULL      |                 |
| product   | varchar(50)   | YES  |      | NULL      |                 |
| quantity  | int             | YES  |    | NULL      |                 |
| total_price | decimal(10,2) | YES  |    | NULL      |                 |
+-------------+---------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE Customers
      -> ADD COLUMN price_per_qnty DECIMAL(10,2);
```

```
Query OK, 0 rows affected (0.00 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW COLUMNS FROM Customers;
+----------------+---------------+------+-----+---------+----------------+
| Field          | Type          | Null | Key | Default | Extra          |
+----------------+---------------+------+-----+---------+----------------+
| cust_id        | int           | NO   | PRI | NULL    | auto_increment |
| cust_name      | varchar(50)   | YES  |     | NULL    |                |
| product        | varchar(50)   | YES  |     | NULL    |                |
| quantity       | int           | YES  |     | NULL    |                |
| total_price    | decimal(10,2) | YES  |     | NULL    |                |
| price_per_qnty | decimal(10,2) | YES  |     | NULL    |                |
+----------------+---------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)

mysql> -- Create view Cust_View (customer ID & name)
mysql> SHOW FULL TABLES WHERE Table_type = 'VIEW';
+--------------------+------------+
| Tables_in_Customer | Table_type |
+--------------------+------------+
| Cust_Synonym       | VIEW       |
+--------------------+------------+
1 row in set (0.00 sec)

mysql> CREATE VIEW Cust_View AS
      -> SELECT cust_id, cust_name
      -> FROM Customers;
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW FULL TABLES WHERE Table_type = 'VIEW';
+--------------------+------------+
| Tables_in_Customer | Table_type |
+--------------------+------------+
| Cust_Synonym       | VIEW       |
| Cust_View          | VIEW       |
+--------------------+------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Cust_View;
+---------+-----------+
| cust_id | cust_name |
+---------+-----------+
|       1 | Alice     |
|       2 | Bob       |
|       3 | Charlie   |
|       4 | Diana     |
|       5 | Ethan     |
+---------+-----------+
5 rows in set (0.00 sec)

mysql> -- Update the view to display customer ID, product, total price
mysql> SELECT * FROM Cust_View;
+---------+-----------+
| cust_id | cust_name |
+---------+-----------+
```

```
|     1 | Alice   |
|     2 | Bob     |
|     3 | Charlie   |
|     4 | Diana   |
|     5 | Ethan   |
+---------+-----------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE OR REPLACE VIEW Cust_View AS
      -> SELECT cust_id, product, total_price
      -> FROM Customers;
```

```
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Cust_View;
+---------+----------+-------------+
| cust_id | product  | total_price |
+---------+----------+-------------+
|     1 | Laptop   |    2000.00 |
|     2 | Phone   |     1500.00 |
|     3 | Tablet   |    500.00 |
|     4 | Monitor  |    400.00 |
|     5 | Keyboard |    250.00 |
+---------+----------+-------------+
5 rows in set (0.00 sec)
```

```
mysql> -- Drop the view
mysql> SHOW FULL TABLES WHERE Table_type = 'VIEW';
```

```
+--------------------+------------+
| Tables_in_Customer | Table_type |
+--------------------+------------+
| Cust_Synonym    | VIEW      |
| Cust_View       | VIEW      |
+--------------------+------------+
2 rows in set (0.00 sec)
```

```
mysql> DROP VIEW Cust_View;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SHOW FULL TABLES WHERE Table_type = 'VIEW';
```

```
+--------------------+------------+
| Tables_in_Customer | Table_type |
+--------------------+------------+
| Cust_Synonym    | VIEW      |
+--------------------+------------+
1 row in set (0.00 sec)
```

```
mysql> -- Create index Cust_index on cust_name
mysql> SHOW INDEX FROM Customers;
```

```
+-----------+------------+----------+--------------+-------------+-----------+--------
-----+----------+--------+------+------------+---------+---------------+---------+----
--------+
```

```
| Table     | Non_unique | Key_name | Seq_in_index | Column_name | Collation |
Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment |
Visible | Expression |
+-----------+------------+----------+--------------+-------------+-----------+--------
-----+----------+--------+------+------------+---------+---------------+---------+----
--------+
| Customers |          0 | PRIMARY  |          1 | cust_id       | A         |
      5 |   NULL |   NULL |   | BTREE     |           |                 | YES       |
NULL        |
+-----------+------------+----------+--------------+-------------+-----------+--------
-----+----------+--------+------+------------+---------+---------------+---------+----
--------+
1 row in set (0.00 sec)
```

```
mysql> CREATE INDEX Cust_index ON Customers (cust_name);
```

```
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW INDEX FROM Customers;
```

```
+-----------+------------+------------+--------------+-------------+-----------+------
-------+----------+--------+------+------------+---------+---------------+---------+--
----------+
| Table     | Non_unique | Key_name   | Seq_in_index | Column_name | Collation |
Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment |
Visible | Expression |
+-----------+------------+------------+--------------+-------------+-----------+------
-------+----------+--------+------+------------+---------+---------------+---------+--
----------+
| Customers |          0 | PRIMARY |          1 | cust_id       | A         |
      5 |   NULL |   NULL |   | BTREE     |           |                 | YES       |
NULL        |
| Customers |          1 | Cust_index |             1 | cust_name   | A
      |           5 |   NULL |   NULL | YES  | BTREE        |           |
      | YES       | NULL      |
+-----------+------------+------------+--------------+-------------+-----------+------
-------+----------+--------+------+------------+---------+---------------+---------+--
----------+
2 rows in set (0.00 sec)
```

```
mysql> -- Drop index Cust_index
mysql> SHOW INDEX FROM Customers;
```

```
+-----------+------------+------------+--------------+-------------+-----------+------
-------+----------+--------+------+------------+---------+---------------+---------+--
----------+
| Table     | Non_unique | Key_name   | Seq_in_index | Column_name | Collation |
Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment |
Visible | Expression |
+-----------+------------+------------+--------------+-------------+-----------+------
-------+----------+--------+------+------------+---------+---------------+---------+--
----------+
| Customers |          0 | PRIMARY |          1 | cust_id       | A         |
      5 |   NULL |   NULL |   | BTREE     |           |                 | YES       |
NULL        |
```

```
| Customers |           1 | Cust_index |              1 | cust_name   | A
      |           5 |   NULL |   NULL | YES  | BTREE        |           |
      | YES       | NULL      |
+-----------+------------+------------+--------------+------------+-----------+------
-------+----------+--------+------+------------+---------+---------------+---------+--
----------+
2 rows in set (0.00 sec)
```

```
mysql> DROP INDEX Cust_index ON Customers;
```

```
Query OK, 0 rows affected (0.00 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> SHOW INDEX FROM Customers;
```

```
+-----------+------------+----------+--------------+------------+-----------+-------
-----+----------+--------+------+------------+---------+---------------+---------+----
--------+
| Table     | Non_unique | Key_name | Seq_in_index | Column_name | Collation |
Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment |
Visible | Expression |
+-----------+------------+----------+--------------+------------+-----------+-------
-----+----------+--------+------+------------+---------+---------------+---------+----
--------+
| Customers |           0 | PRIMARY  |          1 | cust_id       | A         |
      5 |   NULL |   NULL |   | BTREE     |           |                 | YES       |
NULL        |
+-----------+------------+----------+--------------+------------+-----------+-------
-----+----------+--------+------+------------+---------+---------------+---------+----
--------+
1 row in set (0.00 sec)
```

```
mysql> -- Use a table alias / synonym (MySQL alternative)
mysql> SELECT * FROM Customers AS C;
```

```
+---------+-----------+----------+----------+-------------+----------------+
| cust_id | cust_name | product  | quantity | total_price | price_per_qnty |
+---------+-----------+----------+----------+-------------+----------------+
|     1 | Alice   | Laptop   |      2 |   2000.00 |         NULL |
|     2 | Bob     | Phone     |     3 |   1500.00 |         NULL |
|     3 | Charlie   | Tablet   |          1 |   500.00 |          NULL |
|     4 | Diana   | Monitor  |      2 |   400.00 |          NULL |
|     5 | Ethan   | Keyboard |      5 |   250.00 |          NULL |
+---------+-----------+----------+----------+-------------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> SHOW TABLES;
```

```
+--------------------+
| Tables_in_Customer |
+--------------------+
| Cust_Synonym    |
| Customers       |
| customers       |
+--------------------+
3 rows in set (0.00 sec)
```

```
mysql> RENAME TABLE Customers TO Customers_New;
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW TABLES;
+--------------------+
| Tables_in_Customer |
+--------------------+
| Cust_Synonym    |
| Customers_New   |
| customers       |
+--------------------+
3 rows in set (0.00 sec)

mysql> -- Truncate table
mysql> SELECT * FROM Customers_New;
+---------+-----------+----------+----------+-------------+----------------+
| cust_id | cust_name | product  | quantity | total_price | price_per_qnty |
+---------+-----------+----------+----------+-------------+----------------+
|      1 | Alice    | Laptop   |        2 |    2000.00 |           NULL |
|      2 | Bob      | Phone      |      3 |    1500.00 |           NULL |
|      3 | Charlie   | Tablet   |        1 |    500.00 |           NULL |
|      4 | Diana    | Monitor  |        2 |    400.00 |           NULL |
|      5 | Ethan    | Keyboard |        5 |    250.00 |           NULL |
+---------+-----------+----------+----------+-------------+----------------+
5 rows in set (0.00 sec)

mysql> TRUNCATE TABLE Customers_New;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Customers_New;
Empty set (0.00 sec)

mysql> SHOW TABLES;
+--------------------+
| Tables_in_Customer |
+--------------------+
| Cust_Synonym    |
| Customers_New   |
| customers       |
+--------------------+
3 rows in set (0.00 sec)
mysql> DROP TABLE Customers_New;
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW TABLES;
+--------------------+
| Tables_in_Customer |
+--------------------+
| Cust_Synonym    |
| customers       |
+--------------------+
2 rows in set (0.00 sec)
```